```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load Titanic dataset from local CSV file
df = pd.read_csv("titanic.csv")

# ----------------------------------
# 1. Display first 5 rows
# ----------------------------------
print("First 5 rows:")
print(df.head())

# ----------------------------------
# 2. Basic information
# ----------------------------------
print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nData Types:")
print(df.dtypes)

# ----------------------------------
# 3. Check missing values
# ----------------------------------
print("\nMissing Values:")
print(df.isnull().sum())

# ----------------------------------
# 4. Summary statistics
# ----------------------------------
print("\nSummary Statistics:")
print(df.describe())

# ----------------------------------
# 5. Histogram
# ----------------------------------
plt.figure(figsize=(8, 5))
plt.hist(df["Age"].dropna(), bins=20, edgecolor="black")
plt.title("Distribution of Passenger Age")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")
plt.show()

# ----------------------------------
# 6. Fare Histogram
```

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# ----------------------------------
plt.figure(figsize=(8, 5))
plt.hist(df["Fare"].dropna(), bins=20, edgecolor="black")
plt.title("Distribution of Fare")
plt.xlabel("Fare")
plt.ylabel("Number of Passengers")
plt.show()

# ----------------------------------
# 7. Scatter Plot
# ----------------------------------
plt.figure(figsize=(8, 5))
plt.scatter(df["Age"], df["Fare"], alpha=0.5)
plt.title("Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()

# ----------------------------------
# 8. Box Plot - Age
# ----------------------------------
plt.figure(figsize=(8, 5))
plt.boxplot(df["Age"].dropna())
plt.title("Box Plot of Age")
plt.ylabel("Age")
plt.show()

# ----------------------------------
# 9. Box Plot - Fare
# ----------------------------------
plt.figure(figsize=(8, 5))
plt.boxplot(df["Fare"].dropna())
plt.title("Box Plot of Fare")
plt.ylabel("Fare")
plt.show()

# ----------------------------------
# 10. Survival Count
# ----------------------------------
survival_counts = df["Survived"].value_counts()

plt.figure(figsize=(7, 5))
plt.bar(["Not Survived", "Survived"], survival_counts)
plt.title("Titanic Survival Count")
plt.xlabel("Survival Status")
plt.ylabel("Number of Passengers")
plt.show()

# ----------------------------------
```

```
# 11. Survival by Gender
# ----------------------------------
gender_survival = pd.crosstab(df["Sex"], df["Survived"])

gender_survival.plot(
    kind="bar",
    figsize=(8, 5)
)

plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.xticks(rotation=0)
plt.legend(["Not Survived", "Survived"])
plt.show()

# ----------------------------------
# 12. Survival by Passenger Class
# ----------------------------------
class_survival = pd.crosstab(df["Pclass"], df["Survived"])

class_survival.plot(
    kind="bar",
    figsize=(8, 5)
)

plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.xticks(rotation=0)
plt.legend(["Not Survived", "Survived"])
plt.show()
```